

Database Schema Design Specs

BOOK STORE MANAGEMENT SYSTEM

1. Data Modeling Architecture

This layout outlines the key document schemas managed over the Mongoose driver to map structured records into MongoDB collections safely.

2. Schemas & Collection Layouts

2.1 Book Collection Schema (/api/books)

```
{
  "_id": "ObjectId (Auto-Generated)",
  "title": "String",
  "description": "String",
  "category": "String",
  "trending": "Boolean",
  "coverImage": "String",
  "oldPrice": "Number",
  "newPrice": "Number",
  "createdAt": "Timestamp"
}
```

2.2 Order Collection Schema (/api/orders)

```
{
  "_id": "ObjectId (Auto-Generated)",
  "name": "String (Customer Full Name)",
  "email": "String",
  "address": {
    "city": "String",
    "state": "String",
    "country": "String",
    "zipcode": "String"
  },
  "phone": "Number",
  "productIds": ["Array of Reference ObjectIds pointing to Book Collection"],
  "totalPrice": "Number",
  "createdAt": "Timestamp"
}
```

2.3 Admin & Analytics Schema Views (/api/admin)

Does not store distinct isolated records, but relies on pipeline aggregation operations to calculate total product listings, extract historical earnings, and format monthly sales arrays for Chart.js rendering pipelines.